output to enable IC2. LED1 is used as a power-on indicator.

The NE555 timer (IC2) is configured in monostable mode. The time period of IC2 is based on resistors R2 and R4 and capacitor C3, which is around five minutes in this case. By changing values of resistors R2 and R4 and/or capacitor C3, you can change the time period of IC2.

Output pin 3 of IC2 is connected to one end of resistors R3 and R8. The other end of resistor R3 is connected to the base of the solenoid driver transistor T1 to open the solenoid valve. The other end of resistor R8 is connected to the base of LED light driver transistor T2 to light up all the white LEDs (LED2 through LED10).

Normally, the IR sensor module output is high. When a person comes in front of the IR module, the sensor's output becomes low and triggers the NE555. Its output goes high based on the timing components. Here R2 + R4 and C3 are the timing components. Once the NE555 is triggered, its output becomes high for around five minutes. It performs two functions during this period.

First, it keeps the solenoid coil energised so that the valve remains open and the water keeps flowing from the tap for around five minutes. Second, it keeps the LED light on, which also remains on for about five minutes.

## Construction and testing

As the circuit is simple, it can be assembled on a general-purpose PCB. After assembling the circuit, enclose it in a suitable box. Install the IR sensor module in front of the hand wash sink.

Before continuing with the working of the circuit, the setup of the circuit may be understood. The IR module is fixed on the wall just above the wash basin in such a way that when no one is in front of it,

**PARTS LIST**

| | |
|---|---|
| *Semiconductors:* | |
| IC1 | - LM7805, 5V voltage regulator |
| IC2 | - NE555 timer |
| T1, T2 | - BD139 NPN transistor |
| BR1 | - 1A bridge rectifier |
| D1 | - 1N4007 rectifier diode |
| LED1 | - 5mm LED (green or red) |
| LED2-LED10 | - 5mm white LED |
| *Resistors (all 1/4-watt, ±5% carbon):* | |
| R1-R3, R8 | - 1-kilo-ohm |
| R4 | - 270-kilo-ohm |
| R5-R7 | - 470-ohm |
| *Capacitors:* | |
| C1 | - 1000µF, 35V electrolytic |
| C2 | - 0.01µF ceramic disk |
| C3 | - 1000µF, 25V electrolytic |
| *Miscellaneous:* | |
| CON1 | - 2-pin connector |
| SV | - 12V solenoid valve normally closed type |
| X1 | - 230V AC primary to 12V AC, 500mA secondary transformer |

IC2 does not trigger. This means that the tap and LED light fitted above the wash basin will normally be in off state when no one is in front of the IR sensor.

When someone comes in front of the IR sensor fitted above the wash basin, IC2 triggers. This activates the solenoid and water starts flowing from the tap, and simultaneously the LED light turns on, for the preset period.

Fix the solenoid valve in its proper water flow direction, which is marked on the valve housing. There is a minimum pressure required for proper functioning of the solenoid valve, for which the water storage tank should be at a height of at least three metres above the solenoid valve. Otherwise, the use of a water pressure pump would be required. EFY

***Bonus.*** You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/make-your-own-touchless-wash-basin*

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